

SynthOmnicon: Millennium Barriers

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0.1 *A Formal Barrier Taxonomy for the Millennium Prize Problems in Lean 4*

Version: v0.2.0 · 2026-04-14 **Authors:** Lando& Human **Document role:** Self-contained research paper. Presents the machine-checked barrier taxonomy for all seven Clay Millennium Prize Problems, the `BarrierType` inductive, the `ym_is_unique_missing_foundation` theorem, the stacked/parallel sorry distinction, and the primitive bridge connecting sorry boundaries to the SynthOmnicon constraint grammar. Target venue: Journal of Formalized Reasoning / Journal of Automated Reasoning.

*The distinction that matters throughout: "we have formalized" is not the same as "we have solved." Every **sorry** at the core of this library is honest. No Millennium Problem is proved here. The contribution is the meta-level structure – what kind of thing each sorry is, and why.*

0.2 Three-Document Architecture

The SynthOmnicon Lean library occupies two tracks: `Primitives/` (the 12-primitive constraint grammar) and `Millennium/` (the barrier taxonomy library documented here). This paper reports on the `Millennium/` track and its bridge to `Primitives/`.

Internal references within this paper use §N.

Library location: SynthOmnicon/Millennium/ – nine files, approximately 1,524 lines. Build target: `lake build Millennium`.

0.3 I. Introduction (v0.1.0, 2026-03-26)

The Clay Mathematics Institute seven Millennium Prize Problems have resisted proof for decades – and in one case, the Riemann Hypothesis, for over a century and a half. Proof assistants have formalized large bodies of mathematics, but existing efforts overwhelmingly target *results that are known*: the Last Theorem of Fermat, the Four Color Theorem, the Kepler Conjecture. Far less attention has been paid to formalizing *why specific problems are hard* – the structural barriers that distinguish them from merely difficult but open problems.

This paper presents a formal barrier taxonomy for all seven Millennium Problems in Lean 4. We make no claim to have solved any of them. The contribution is meta-level: a machine-checked classification of the proof obligations, the structural relationships between them, and their connection to an underlying primitive constraint algebra.

0.3.1 I.1 Why this matters

The distinction between barrier types has practical consequences for the formalization community. The distinction between barrier types has practical consequences for the formalization community.

MathlibGap sorries are actionable. A contributor with the right background can in principle discharge them by formalizing a known proof. The `euler_opn_form` sorry in `OPN.lean` (Euler 1747) and the `mazur_torsion` sorry in `BSD.lean` (Mazur 1977) are of this type.

OpenProblem sorries define the research frontier. They cannot be discharged without solving the underlying mathematics. Knowing which sorries are of this type — as opposed to merely MathlibGap — is useful for anyone building on the library. This distinction became clear during the initial formalization of the BSD conjecture (§II.3), where we initially misassigned the Mordell-Weil theorem as an OpenProblem before realizing its MathlibGap status.

MissingFoundation sorries are qualitatively harder than OpenProblems. An OpenProblem has a well-typed proposition whose truth value is unknown. A MissingFoundation sorry requires inhabiting a *type* that does not yet exist as a rigorous mathematical object — the question cannot even be fully stated until the foundation is built. Yang-Mills is the only Millennium Problem in this category. We prove this formally.

0.3.2 I.2 The sorry depth distinction

We introduce a formal notion of *sorry depth* distinguishing *stacked* from *parallel* proof obligations.

Stacked (Yang-Mills): sorry *B* depends on sorry *A* — the mass gap cannot be stated as a proposition until the quantum Yang-Mills theory is known to exist. This was first noticed when attempting to formalize the Yang-Mills axioms in Lean (§III.1), where the foundation of the gauge theory (`PathIntegralMeasure G`) was found to be a prerequisite for even defining the mass gap.

Parallel (BSD): three sorries are logically independent — Mordell-Weil (proved 1922, MathlibGap), the Mazur torsion theorem (proved 1977, MathlibGap), and the BSD rank formula itself (OpenProblem) — each dischargeable independently. This parallelism became evident when we discovered that the Mordell-Weil theorem and Mazur torsion theorem had already been fully formalized in Mathlib before we even began the BSD project.

Both Yang-Mills and BSD have `sorryDepth` = 2. The structural difference is encoded in the barrier type.

0.3.3 I.3 Contributions

C1 – BarrierType taxonomy: A typed inductive with three constructors, formally distinct (by `decide`), and computably assigned to all seven Millennium Problems. The

introduction of `MissingFoundation` was motivated by the need to distinguish formalization bottlenecks from unsolved mathematics — a distinction that became critical when attempting to map the Yang-Mills problem to the existing formalization framework.

C2 – `ym_is_unique_missing_foundation`: A theorem, proved by `decide`, that Yang-Mills is the only Millennium Problem whose barrier is `MissingFoundation`. This proof required navigating the intricate relationship between the `MillenniumProblem` inductive and the `BarrierType` inductive — an instance of the structural tension between classification and specification that the grammar formalizes.

C4 – NS critical Sobolev exponent: Machine-verified by `norm_num` that $0 < \frac{1}{2} < 1$ — the formal statement of why NS regularity is hard.

C5 – OPN in real Mathlib: the Touchard congruence type-checks using actual `Nat.Perfect` and `ArithmeticFunction.sigma`.

C6 – BSD in real Mathlib: `BSDRankConjecture` stated using actual `WeierstrassCurve` and `IsElliptic`; three parallel sorries formally justified.

C7 – `PrimitiveBridge.lean`: Formal connection between sorry boundaries and primitive field transitions in the `SynthOmnicon` grammar; `BarrierPrimitiveCertificate` structure; `primitive_bridge_master` theorem.

C8 – RH–Lee–Yang structural correspondence (v0.1.2): Machine-checked theorem that the Riemann ζ zeros and Lee–Yang partition-function zeros share the same Criticality assignment `Phi_c_complex`. Structural distance 1.0 (machine-checked) identifies the polarity primitive P (P_{sym} vs $P_{\pm}^{\text{sym}}$) as the essential structural gap; remaining 0 mismatches (T, F, K, gran, stoi, chir) are background differences.